

Credit Card Portfolio Risk Analytics

Project: Credit Card Portfolio Risk Analytics using SQL & Power BI

Dataset: Taiwan Credit Card Default Dataset

Records: 30,000 Customers

Target Variable: Default Payment Next Month

1. Dataset Overview

This project analyzes credit card customer behaviour to identify default risk, segment customers based on financial characteristics, and build an interactive dashboard for portfolio monitoring.

The dataset contains demographic information, historical repayment behaviour, monthly bill statements, monthly repayments, and engineered risk features created during the SQL phase of the project.

2. Original Variables

Column	Data Type	Description
ID	Integer	Unique customer identifier
LIMIT_BAL	Integer	Credit limit assigned to the customer (NT\$)
SEX	Integer	Gender (1 = Male, 2 = Female)
EDUCATION	Integer	Education level (1 = Graduate School, 2 = University, 3 = High School, 4 = Others)
MARRIAGE	Integer	Marital status (1 = Married, 2 = Single, 3 = Others)
AGE	Integer	Customer age (years)
PAY_0	Integer	Repayment status in September 2005
PAY_2	Integer	Repayment status in August 2005
PAY_3	Integer	Repayment status in July 2005
PAY_4	Integer	Repayment status in June 2005
PAY_5	Integer	Repayment status in May 2005
PAY_6	Integer	Repayment status in April 2005
BILL_AMT1	Currency	September bill amount
BILL_AMT2	Currency	August bill amount

BILL_AMT3	Currency	July bill amount
BILL_AMT4	Currency	June bill amount
BILL_AMT5	Currency	May bill amount
BILL_AMT6	Currency	April bill amount
PAY_AMT1	Currency	September payment
PAY_AMT2	Currency	August payment
PAY_AMT3	Currency	July payment
PAY_AMT4	Currency	June payment
PAY_AMT5	Currency	May payment
PAY_AMT6	Currency	April payment
default payment next month	Binary	Target variable (0 = No Default, 1 = Default)

3. Variable Coding

Gender

Code	Meaning
1	Male
2	Female

Education

Code	Meaning
1	Graduate School
2	University
3	High School
4	Others

Marital Status

Code	Meaning
1	Married
2	Single
3	Others

Repayment Status (PAY_0 – PAY_6)

Value	Meaning
-1	Paid duly
0	Revolving credit / No delay
1	Payment delayed by 1 month
2	Payment delayed by 2 months
3	Payment delayed by 3 months
4	Payment delayed by 4 months
5	Payment delayed by 5 months
6	Payment delayed by 6 months
7	Payment delayed by 7 months
8	Payment delayed by 8 months
9	Payment delayed by 9 months or more

4. Engineered Features (SQL)

The following variables were created during the SQL data preparation phase to support portfolio risk analysis.

Variable	Description	Formula / Logic
total_bill	Total outstanding bills across six months	SUM(BILL_AMT1–BILL_AMT6)
total_payment	Total repayments across six months	SUM(PAY_AMT1–PAY_AMT6)

payment_ratio	Proportion of bills repaid	$\text{total_payment} \div \text{total_bill}$
max_delay	Maximum repayment delay recorded	$\text{MAX}(\text{PAY_0} \dots \text{PAY_6})$
age_group	Customer age segment	Based on AGE
credit_band	Credit limit category	Based on LIMIT_BAL
risk_score	Composite repayment risk indicator	Derived using repayment delays and payment behaviour
risk_category	Overall customer risk classification	Low Risk / Medium Risk / High Risk

5. Derived Variable Definitions

Age	Group
21–30	Young Adults
31–40	Early Career
41–50	Mid-Career
51–60	Senior
60+	Elderly

Credit Band

Credit Limit (NT\$)	Band
< 50,000	Low
50,000 – 149,999	Medium
150,000 – 299,999	High
≥ 300,000	Premium

Risk Category

Category	Interpretation
Low Risk	Strong repayment behaviour with low delinquency
Medium Risk	Moderate repayment delays requiring monitoring
High Risk	Persistent repayment issues and elevated probability of default

6. Business Measures Used in Power BI

Measure	Description
Total Customers	Total number of customers
Total Defaulters	Customers who defaulted (Target = 1)
Default Rate	Percentage of customers who defaulted
Portfolio Exposure	Total outstanding bill amount
Average Credit Limit	Mean customer credit limit
Average Payment Ratio	Mean payment ratio across customers
High Risk Customers	Number of customers classified as High Risk
Average Risk Score	Mean calculated risk score

7. Data Source

Dataset: Default of Credit Card Clients (Taiwan)